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KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY AVINASHI ROAD, ARASUR, COIMBATORE – 641407

FOOD DELIVERY DELAY ANALYSIS AND PREDICTION

PROJECT REPORT

Submitted by

ANUSKA GUPTA (24AD127)

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ABSTRACT

Food delivery services depend on timely order fulfillment to improve customer satisfaction and operational efficiency. This project explores a real-world food delivery dataset containing 45,593 records and 20 original fields to understand the factors associated with delivery duration and develop a prediction model.

The study uses Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and TensorFlow, for data cleaning, statistical analysis, visualization, preprocessing, and modeling. Exploratory analysis examines delivery-time distributions, outliers, geographical distance, traffic density, weather conditions, vehicle type, and other order-related variables.

A neural network is trained and evaluated alongside a mean-prediction baseline and a Random Forest regression model. The Random Forest model is selected using validation mean absolute error and achieves a test MAE of 3.16 minutes, RMSE of 4.00 minutes, and R^2 of 0.8140. The findings demonstrate the usefulness of data exploration and regression modeling for estimating delivery duration.

Keywords: Exploratory Data Analysis, Food Delivery, Data Visualization, Regression, Deep Learning, Random Forest, Delivery-Time Prediction

1. INTRODUCTION

1.1 Background

Online food delivery platforms connect customers, restaurants, and delivery personnel. Delivery time is an important operational measure because it affects customer satisfaction, order planning, and the allocation of delivery resources.

Delivery duration may be associated with several factors, including the distance between a restaurant and a customer, traffic conditions, weather, delivery-person characteristics, vehicle type, multiple orders, and the time of order placement.

Understanding these relationships requires systematic examination of the available data. Exploratory Data Analysis (EDA) helps identify patterns, distributions, missing values, unusual observations, and relationships between variables before predictive modeling is performed.

This project applies EDA and visualization techniques to a food delivery dataset and develops regression models to estimate delivery duration.

1.2 Problem Statement

Food delivery duration varies from one order to another. Without a data-driven estimate, it can be difficult to anticipate unusually long deliveries and plan operations efficiently.

The problem addressed in this project is to investigate the characteristics associated with delivery duration and build a regression model that predicts the time taken to deliver an order using available order, delivery, geographical, and environmental features.

1.3 Objectives

The main objectives are:

1. To load and understand a real-world dataset containing more than 15 fields.
2. To identify and handle missing values, invalid entries, duplicates, and unusual observations.
3. To perform univariate, bivariate, and multivariate statistical analysis.
4. To create and interpret meaningful visualizations.
5. To engineer useful features from date, time, and geographical coordinates.
6. To encode categorical features and scale numerical features where appropriate.
7. To split the dataset into training, validation, and test sets.
8. To develop a deep-learning regression model using a multilayer perceptron.
9. To compare the neural network with a baseline and a Random Forest regressor.
10. To evaluate model performance using regression metrics and error visualizations.

1.4 Scope of the Project

The project focuses on understanding and predicting delivery duration from the information available in the dataset. It includes exploratory analysis, preprocessing, feature engineering, model training, model comparison, and final evaluation.

The project is an offline analytical study. It does not represent a live food delivery platform, and the predictions should not be interpreted as guaranteed delivery times.

2. DATASET DESCRIPTION

2.1 Dataset Overview

The project uses a food delivery dataset loaded from `train.csv`.

Property	Description
Dataset file	`train.csv`
Original number of records	45,593
Original number of columns	20
Final engineered dataset	45,593 rows × 19 columns
Prediction target	`Time_taken(min)`
Problem type	Supervised regression
Data characteristics	Numerical, categorical, date, time, and geographical fields
Data analysis environment	Python in Google Colab / Jupyter Notebook

The original dataset contains 20 fields. During feature engineering, identifier and intermediate date/time fields are removed, while new date, time, and distance features are created. The resulting dataset contains 19 fields, including the target, and 18 predictor fields.

Dataset source: [Food Delivery Dataset]

2.2 Description of Original Fields

The original dataset contains the following fields:

Field	Description
`ID`	Unique order identifier
`Delivery_person_ID`	Identifier of the delivery person
`Delivery_person_Age`	Age of the delivery person
`Delivery_person_Ratings`	Rating associated with the delivery person
`Restaurant_latitude`	Restaurant latitude
`Restaurant_longitude`	Restaurant longitude
`Delivery_location_latitude`	Customer delivery latitude
`Delivery_location_longitude`	Customer delivery longitude
`Order_Date`	Date on which the order was placed
`Time_Orderd`	Time at which the order was placed

`Time_Order_picked`	Time at which the order was picked up
`Weatherconditions`	Recorded weather condition
`Road_traffic_density`	Recorded traffic category
`Vehicle_condition`	Vehicle condition indicator
`Type_of_order`	Type or category of food order
`Type_of_vehicle`	Vehicle used for delivery
`multiple_deliveries`	Number/category of deliveries handled together
`Festival`	Festival indicator
`City`	City classification
`Time_taken(min)`	Delivery duration in minutes; prediction target

2.3 Target Variable

The target variable is `Time_taken(min)`, which represents the delivery duration in minutes.

This is a continuous numerical prediction problem. Therefore, regression algorithms and metrics such as Mean Absolute Error, Root Mean Squared Error, and the coefficient of determination are suitable for evaluation.

3. DATA UNDERSTANDING AND PREPROCESSING

3.1 Initial Data Inspection

The dataset was loaded using Pandas. Initial inspection included:

- Dataset dimensions.
- Column names and data types.
- Numerical descriptive statistics.
- Categorical variable summaries.
- Missing-value analysis.
- Duplicate-record detection.
- Target-variable validation.

The original dataset contained 45,593 rows and 20 columns. No exact duplicate rows were identified, and no rows with missing target values were removed during the cleaning stage.

3.2 Missing-Value Handling

Missing values were investigated in both numerical and categorical fields. Text values representing missing data, including blank strings and placeholder values such as `nan`, `null`, `none`, and `not available`, were normalized where applicable.

After feature engineering, the quality report identified the following missing values:

Field	Missing values	Percentage
`Delivery_person_Age`	1,854	4.07%
`Delivery_person_Ratings`	1,908	4.18%
`Weatherconditions`	616	1.35%
`Road_traffic_density`	601	1.32%
`multiple_deliveries`	993	2.18%
`Festival`	228	0.50%
`City`	1,200	2.63%
`Order_Hour`	1,731	3.80%
`Distance_km`	431	0.95%

The displayed quality report showed zero missing values in the target variable and zero duplicate rows.

The 616 missing weather values were excluded from the weather-specific summary analysis rather than being treated as a valid weather category.

For predictive modeling, missing numerical values were imputed using the median and missing categorical values using the most frequent category. The imputers were fitted on the training data to avoid learning preprocessing parameters from the validation and test sets.

3.3 Duplicate and Target Validation

The initial duplicate check found zero exact duplicate rows. The final quality check also reported:

- Duplicate rows remaining: 0
- Target missing values: 0
- Total fields after feature engineering: 19
- Predictor fields: 18

These checks helped ensure that the final modeling dataset was structurally consistent.

3.4 Outlier Analysis

A boxplot and the interquartile range (IQR) method were used to examine unusual delivery durations.

The observed target statistics were:

Statistic	Value
Mean	26.29 minutes
Median	26 minutes
Standard deviation	9.38 minutes
Skewness	0.486
IQR-based upper fence	51.5 minutes
Potential outliers	270
Potential outliers as a percentage	0.59%

The delivery-time distribution was moderately right-skewed. A small number of deliveries took considerably longer than the central range.

These observations were not automatically removed because unusually long delivery durations may represent genuine delays that are important to the prediction problem.

3.5 Feature Engineering

Feature engineering transformed existing information into variables that could be used by the models.

Day of the week

The `Order_Date` field was converted into a day-of-week feature named `Order_DayOfWeek`.

Month

The month was extracted from the order date and stored as `Order_Month`.

Order hour

The order time was converted into a numerical hour feature named `Order_Hour`. Unparseable or missing times were treated as missing values for subsequent preprocessing.

Geographical distance

Restaurant and delivery-location coordinates were used to estimate straight-line distance in kilometres using the Haversine formula. The resulting feature was named `Distance_km`.

The estimated distance is not the same as the actual road distance, which may be longer because of roads, traffic restrictions, and route selection.

Distances exceeding the selected plausibility threshold of 50 km were flagged as missing rather than used as valid distance measurements. The final distance-quality summary contained 45,162 valid distance values, with a mean of approximately 9.74 km and a maximum of approximately 20.97 km.

Removal of unsuitable fields

Identifier fields and intermediate date/time fields were removed from the modeling feature set where appropriate. The pickup-wait feature, `Pickup_Wait_Min`, was excluded from the target-related correlation analysis and modeling features to avoid potential information leakage.

3.6 Encoding and Standardization

Categorical fields were converted into numerical model inputs using one-hot encoding. Numerical features were standardized where required by the neural network.

The preprocessing workflow used separate numerical and categorical transformations:

- Numerical features: median imputation and standardization.
- Categorical features: most-frequent-value imputation and one-hot encoding.
- Unknown categories: handled through the encoder configuration.

The preprocessing transformations were fitted using the training data and then applied to validation and test data.

3.7 Dataset Splitting

The dataset was divided into three subsets:

Subset	Number of records	Percentage
Training	31,915	70%
Validation	6,839	15%
Testing	6,839	15%

Total	**45,593**	**100%**

The training set was used to fit the models. The validation set was used for model comparison and selection. The test set was reserved for the final evaluation of the selected model.

4. EXPLORATORY DATA ANALYSIS AND VISUALIZATION

The EDA stage examined individual variables, relationships between variables, and patterns relevant to food delivery operations.

Each visualization was selected to answer a particular analytical question and was interpreted in the context of the observed data.

4.1 Visualization 1: Distribution of Delivery Time

Visualization type: Histogram with distribution analysis.

What it shows:

The histogram displays how frequently different delivery durations occur.

Why it was created:

It helps identify typical delivery times, spread, skewness, and the presence of unusually long deliveries.

Observed findings:

The mean delivery duration was 26.29 minutes, while the median was 26 minutes. The standard deviation was 9.38 minutes, and the skewness was 0.486.

Interpretation:

The mean and median are close, suggesting that delivery duration is centred around the mid-twenties in minutes. The positive skewness indicates a longer right tail, meaning that some orders take considerably longer than typical orders.



4.2 Visualization 2: Delivery-Time Boxplot and Outliers

Visualization type: Boxplot.

What it shows:

The boxplot summarizes the median, quartiles, spread, and potential outliers of delivery duration.

Why it was created:

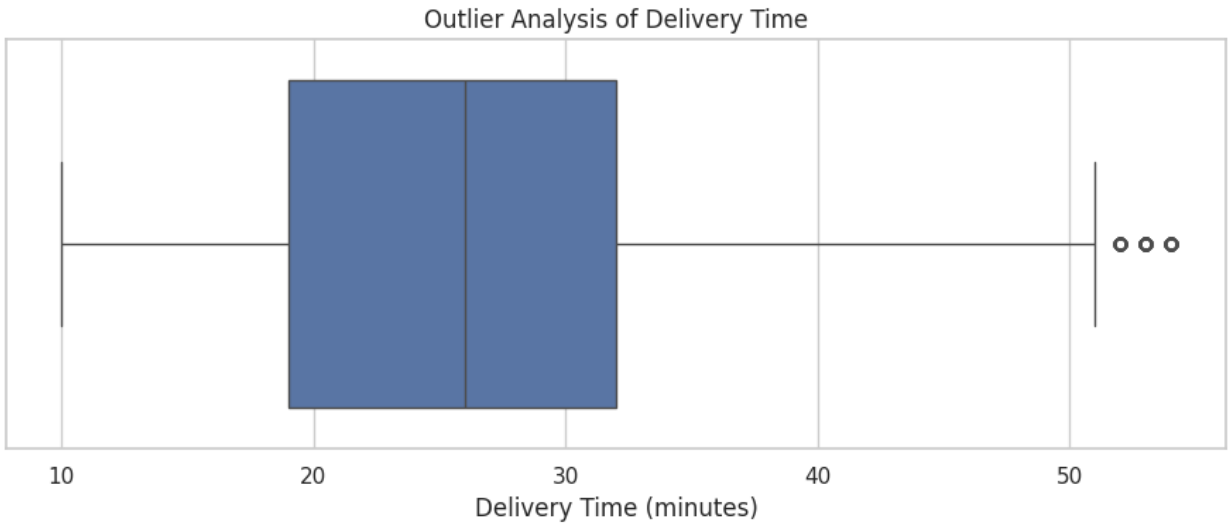
It helps identify unusually short or long deliveries and understand the variability of the target variable.

Observed findings:

The IQR-based upper fence was 51.5 minutes, and 270 observations, approximately 0.59% of the dataset, were identified as potential outliers.

Interpretation:

Some orders take much longer than the central delivery-time range. Such observations may reflect operational delays rather than incorrect records. Therefore, they were examined rather than automatically deleted.



4.3 Visualization 3: Correlation Heatmap

Visualization type: Numerical correlation heatmap.

What it shows:

The heatmap displays pairwise Pearson correlation coefficients among numerical variables.

Why it was created:

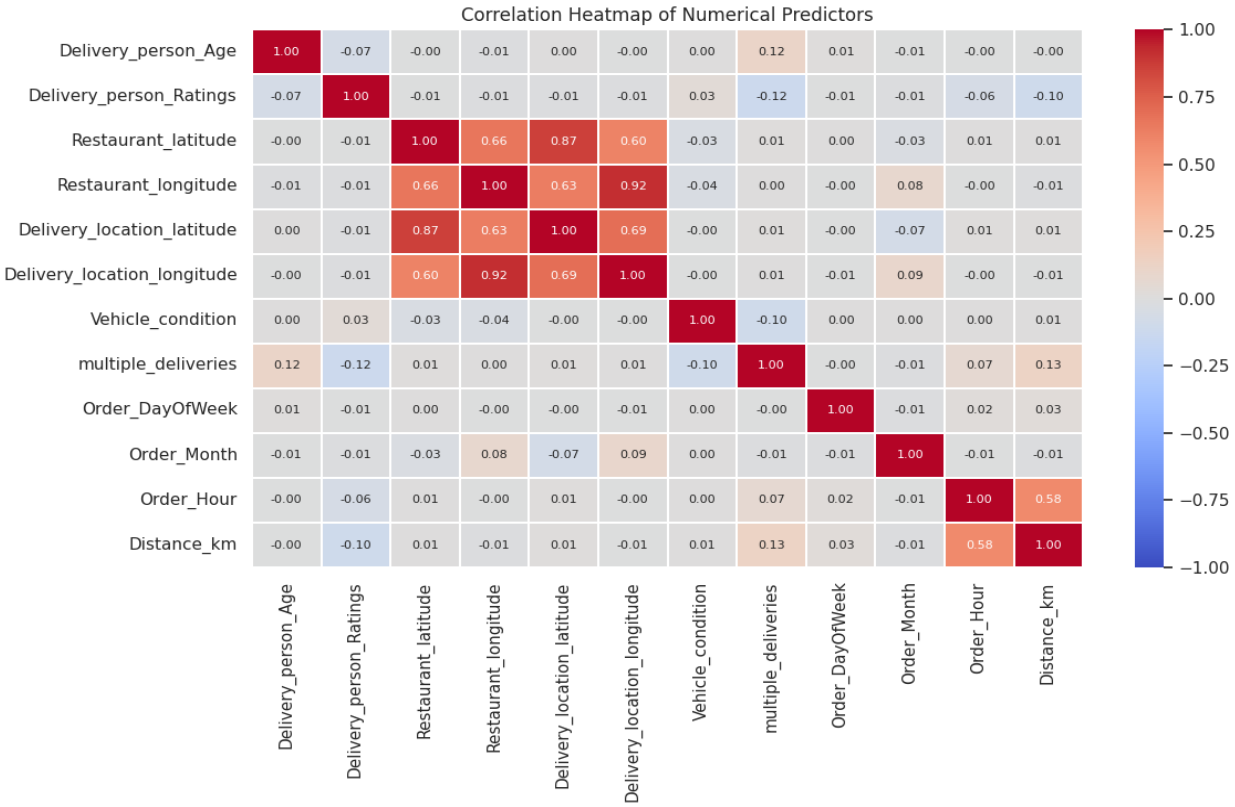
It helps identify numerical variables that have stronger linear relationships with one another and supports feature analysis.

Observed findings:

The analysis indicates that relationships between delivery time and individual numerical features vary in direction and strength. The feature-correlation analysis also highlights variables that deserve further investigation.

Interpretation:

Positive correlation indicates that two variables tend to increase together, while negative correlation indicates an inverse linear relationship. Correlation does not establish causation and may not reveal nonlinear relationships.



4.4 Visualization 4: Delivery Distance versus Delivery Time

Visualization type: Scatterplot.

What it shows:

The scatterplot compares estimated restaurant-to-customer straight-line distance with delivery duration.

Why it was created:

It helps assess whether longer distances tend to be associated with longer delivery times and whether the relationship is linear or variable.

Observed findings:

The distance feature has a positive Pearson correlation of approximately 0.322 with delivery duration in the cleaned numerical feature analysis.

Interpretation:

The positive association is consistent with longer trips often requiring more time. However, the spread of points indicates that distance alone cannot explain delivery duration. Traffic, weather, order handling, and other operational factors may also contribute.



4.5 Visualization 5: Traffic Density versus Delivery Time

Visualization type: Boxplot by traffic category.

What it shows:

The plot compares the distribution of delivery duration across traffic categories.

Why it was created:

Traffic conditions may be associated with delays, and comparing categories helps identify differences in typical duration and variability.

Observed findings:

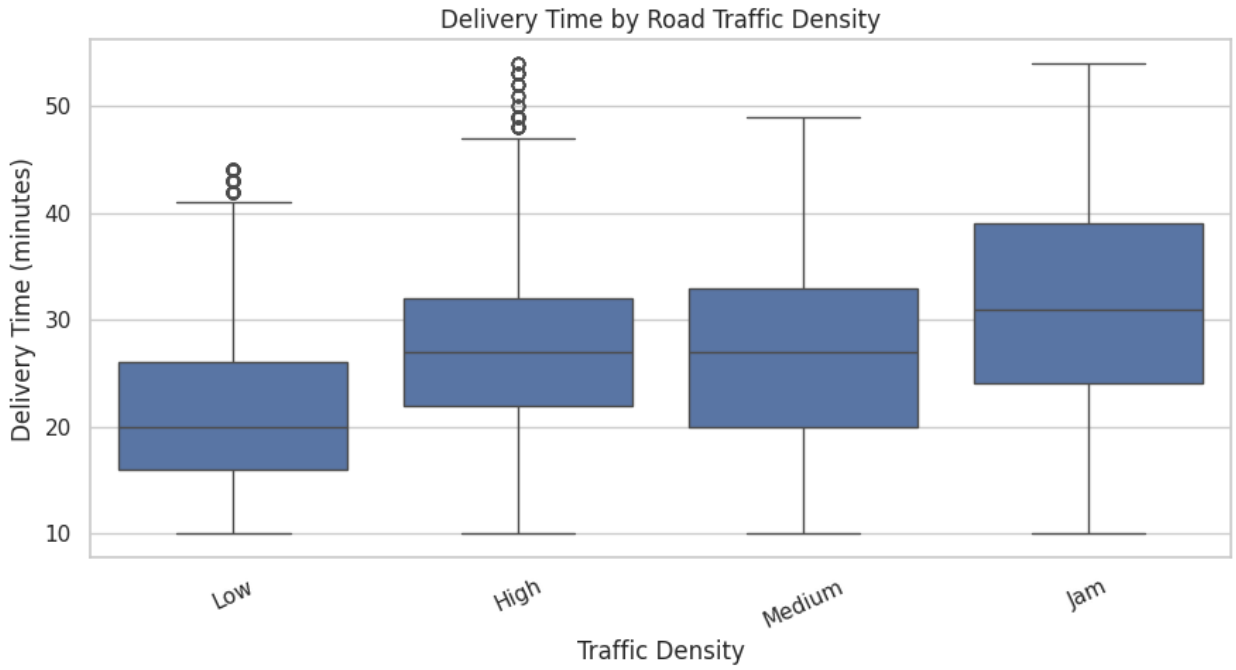
The grouped mean delivery times were:

Traffic category	Mean delivery time
Low	21.27 minutes
High	27.24 minutes

Medium	26.70 minutes
Jam	31.18 minutes

Interpretation:

The Jam category has the highest mean delivery duration among the listed traffic categories, while Low has the lowest. This pattern is consistent with heavier traffic being associated with longer deliveries. However, these are observational group comparisons and do not prove that traffic alone caused the differences.



4.6 Visualization 6: Weather Conditions versus Delivery Time

Visualization type: Boxplot grouped by weather condition.

What it shows:

The plot compares delivery-time distributions across six valid weather categories.

Why it was created:

It helps explore whether delivery durations differ among recorded weather conditions.

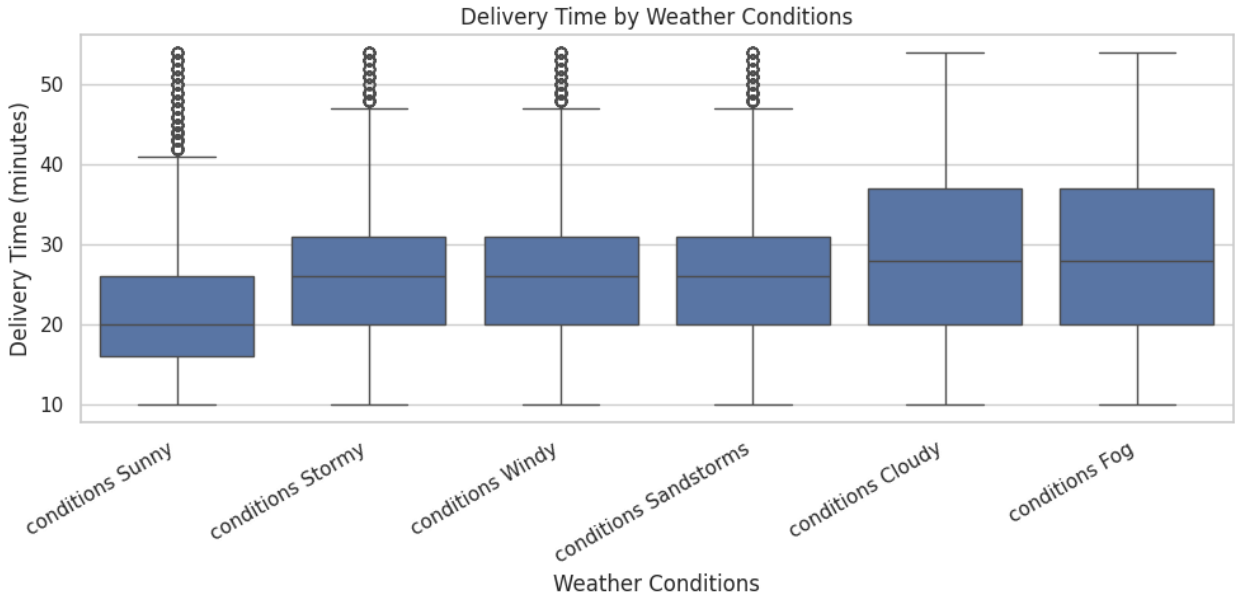
Observed findings:

Weather condition	Count	Mean (minutes)	Median (minutes)
Cloudy	7,536	28.92	28
Fog	7,654	28.92	28
Sandstorms	7,495	25.88	26
Stormy	7,586	25.87	26
Sunny	7,284	21.86	20
Windy	7,422	26.12	26

A total of 616 records with missing weather information were excluded from this summary.

Interpretation:

Sunny conditions have the lowest observed mean and median delivery durations among the listed categories, while Cloudy and Fog have higher observed means. These differences indicate an association between the recorded weather category and delivery duration in this dataset. They do not establish a causal effect, and other factors may influence the observed differences.



4.7 Visualization 7: Vehicle Type versus Delivery Time

Visualization type: Boxplot grouped by vehicle type.

What it shows:

The plot compares delivery-time distributions across vehicle categories.

Why it was created:

Different vehicle types may be used under different delivery conditions, and their observed delivery durations can be compared.

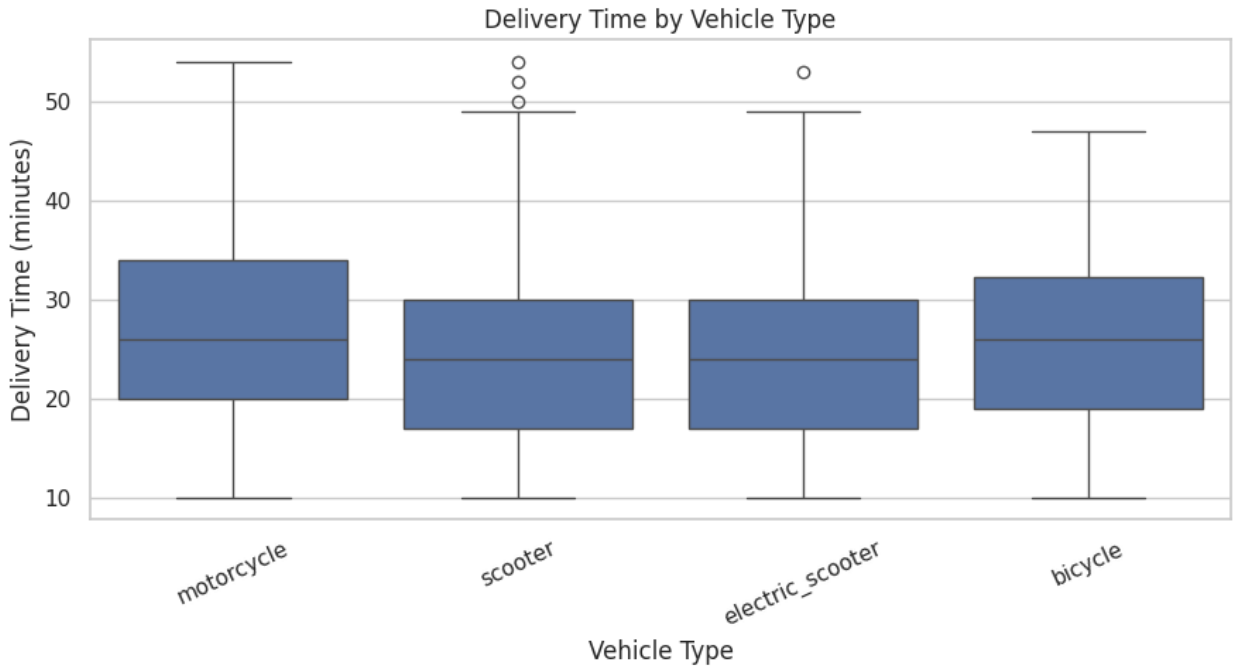
Observed findings:

The notebook's grouped statistics report the following mean delivery times:

Vehicle type	Mean delivery time
Bicycle	26.43 minutes
Electric scooter	24.47 minutes
Motorcycle	27.61 minutes
Scooter	24.48 minutes

Interpretation:

Motorcycles have the highest observed mean delivery duration among these categories, while electric scooters and scooters have lower means. The bicycle category contains only 68 observations, so its mean should be interpreted cautiously. These comparisons do not establish that vehicle type alone determines delivery time.



4.8 Additional Statistical Analysis: City, Festival, and Multiple Deliveries

The project also compares delivery duration across city classifications, festival status, and multiple-delivery categories.

City classification

The observed mean delivery times were approximately:

City classification	Mean delivery time
Metropolitian	27.32 minutes
Semi-Urban	49.73 minutes
Urban	22.98 minutes

The Semi-Urban category has a much higher observed mean than the other categories. However, its sample size is only 164, so the result should be interpreted cautiously.

Festival status

Festival status	Mean delivery time
Yes	45.52 minutes
No	25.98 minutes

The Festival = Yes category has a higher observed mean. The number of observations in that group is 896, compared with a substantially larger No group. The difference may reflect several factors and should not be interpreted as proof of a causal festival effect.

Multiple deliveries

Multiple-delivery category	Mean delivery time
0	22.88 minutes
1	26.86 minutes
2	40.45 minutes
3	47.82 minutes

The observed mean delivery duration increases across the listed multiple-delivery categories. This pattern suggests that handling more deliveries together may be associated with longer delivery durations.

5. DEEP-LEARNING MODEL DEVELOPMENT

5.1 Problem Formulation

Delivery-time prediction is treated as a supervised regression problem.

Let (X) represent the available predictor features and (y) represent delivery duration in minutes. The objective is to learn a function that estimates delivery duration from the input features:

[

\hat{y}=f(X)

]

The target is continuous, so the model produces a numerical prediction rather than a class label.

5.2 Multilayer Perceptron Architecture

A Multilayer Perceptron (MLP) neural network was implemented using TensorFlow and Keras.

The architecture consists of:

1. An input layer that accepts the processed feature vector.
2. A fully connected hidden layer with 64 neurons and ReLU activation.
3. A dropout layer with a dropout rate of 0.20.
4. A second fully connected hidden layer with 32 neurons and ReLU activation.
5. An output layer with one neuron and linear activation to predict delivery time.

The ReLU activation allows the network to model nonlinear relationships. Dropout is used as a regularization technique to reduce over-reliance on individual neurons.

5.3 Training Configuration

Parameter	Configuration

Framework	TensorFlow / Keras
Model type	Multilayer Perceptron regression
Hidden layer 1	64 neurons, ReLU
Dropout	0.20
Hidden layer 2	32 neurons, ReLU
Output layer	1 neuron, linear
Optimizer	Adam
Learning rate	0.001
Loss function	Mean Squared Error
Training metric	Mean Absolute Error
Maximum epochs	60
Batch size	64
Early stopping	Validation loss
Early-stopping patience	8 epochs
Best weights	Restored after early stopping

Early stopping monitors validation loss and can stop training when further epochs do not improve the monitored metric. The notebook completed 60 epochs in the recorded run.

5.4 Baseline and Random Forest Models

The neural network was compared with two additional regression approaches.

Mean-prediction baseline

A `DummyRegressor` using the mean strategy was used as a baseline. It predicts a constant value based on the training target and provides a reference against which more complex models can be evaluated.

Random Forest regressor

A Random Forest regressor was trained using the processed training data.

Its main configuration was:

- Number of trees: 150
- Minimum samples per leaf: 2
- Random state: 42
- Parallel processing: enabled

The model comparison used validation MAE. The model with the lowest validation MAE was selected for final evaluation.

5.5 Model Selection

The validation dataset was used to compare the baseline, Random Forest, and neural network. The Random Forest model was selected because it achieved the lowest validation MAE in the recorded notebook run.

The test dataset was not used to select the winning model. It was reserved for evaluating the selected model after the validation-based selection process.

6. MODEL EVALUATION AND RESULT VISUALIZATION

6.1 Evaluation Metrics

Three metrics were used to evaluate the selected regression model.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted delivery times. It is expressed in minutes, making it straightforward to interpret.

[

MAE=\frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|

]

A lower MAE indicates smaller average absolute prediction errors.

Root Mean Squared Error (RMSE)

RMSE is the square root of the mean squared prediction error. It penalizes larger errors more strongly than MAE.

[

RMSE=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}

]

A lower RMSE indicates smaller squared prediction errors overall.

Coefficient of Determination ((R^2))

The R^2 score compares the model's predictions with a baseline that predicts the mean target value.

An R^2 value closer to 1 indicates that the model explains more of the variation in the target on the evaluated data.

6.2 Training Loss versus Epoch

Visualization type: Line chart of training and validation loss.

What it shows:

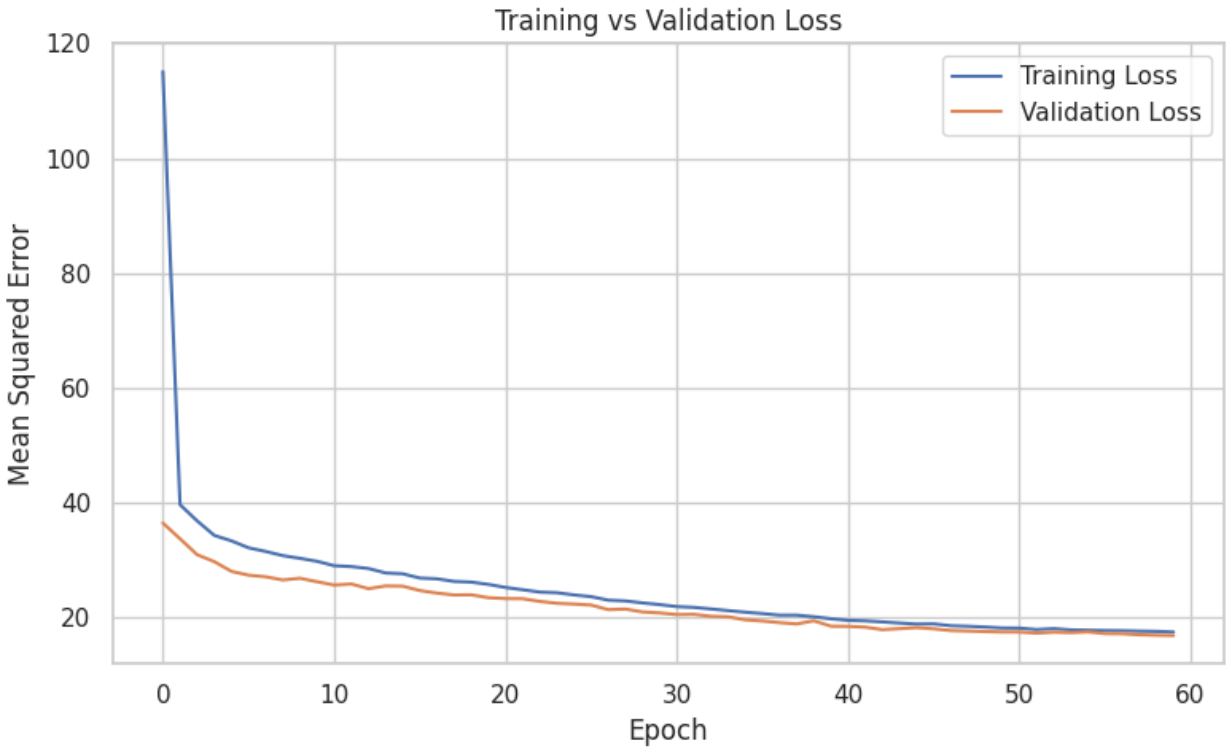
The chart displays how mean squared error changes over training epochs for the training and validation datasets.

Why it was created:

It helps evaluate the learning process and identify possible overfitting.

Interpretation:

A reduction in both training and validation loss indicates that the model is learning useful patterns. A sustained increase in validation loss while training loss continues to decrease may indicate overfitting. Early stopping and restoration of the best weights help control this risk.



6.3 Mean Absolute Error versus Epoch

Visualization type: Line chart of training and validation MAE.

What it shows:

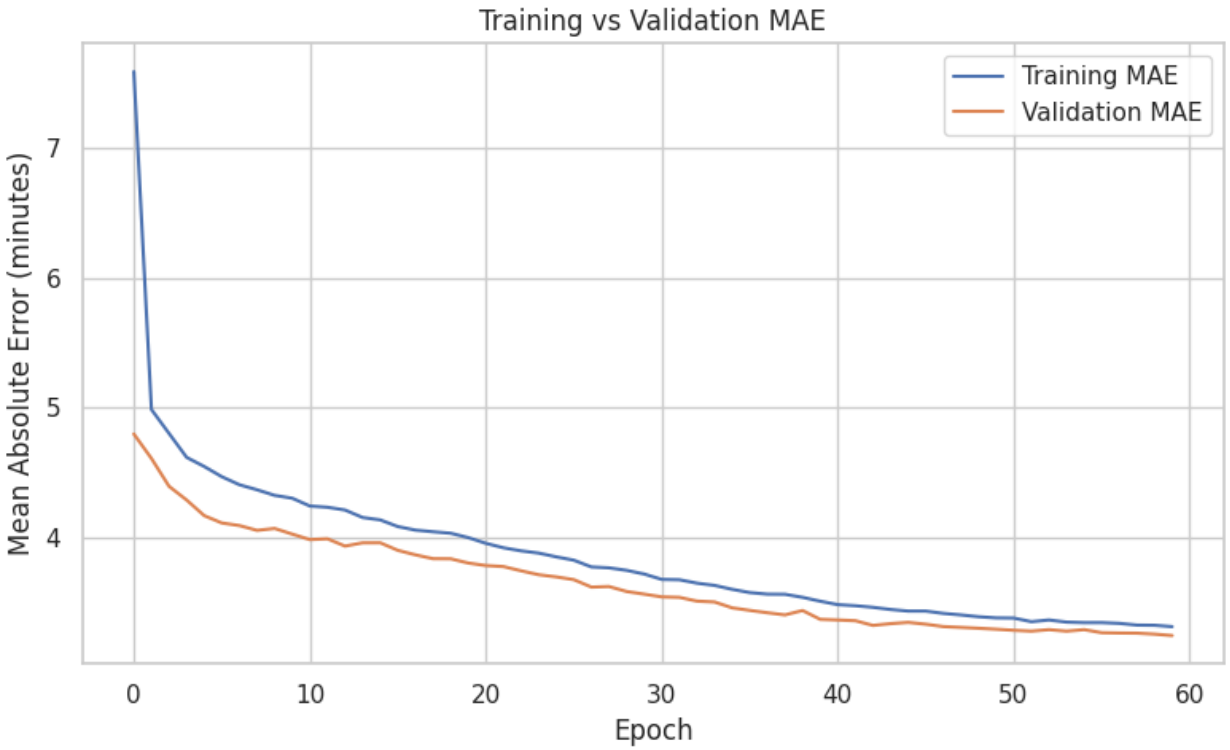
The chart displays the change in average absolute prediction error over the training epochs.

Why it was created:

It helps track the model's prediction error during training and compare performance on training and validation data.

Interpretation:

Lower MAE indicates more accurate predictions on average. Comparing training and validation curves helps assess whether the model generalizes to unseen validation observations.



6.4 Actual versus Predicted Delivery Time

Visualization type: Scatterplot of actual versus predicted values.

What it shows:

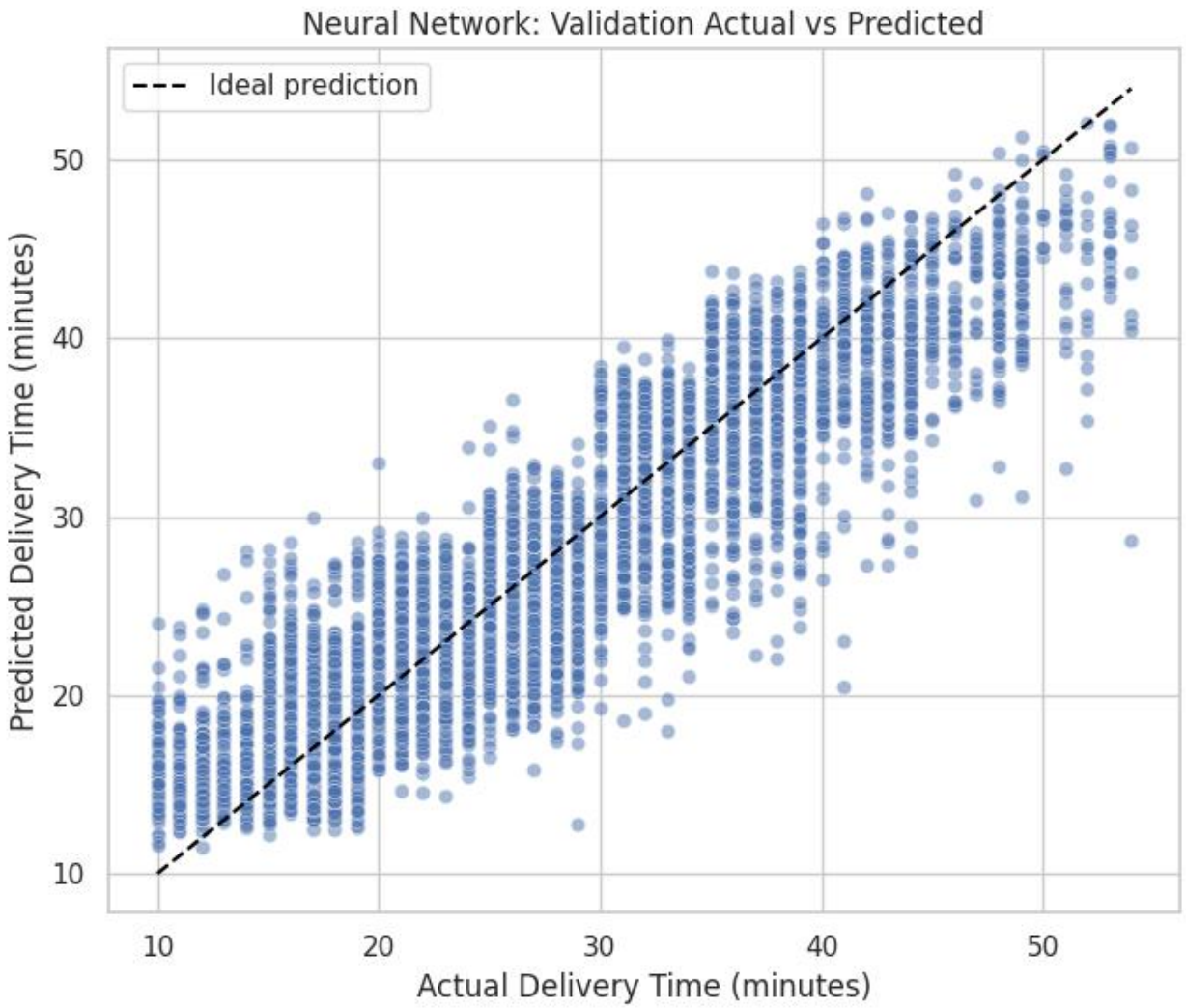
Each point compares an actual delivery duration with the model's corresponding predicted duration.

Why it was created:

It provides a visual assessment of prediction accuracy and systematic errors across the target range.

Interpretation:

Points near the diagonal reference line indicate close agreement between predicted and actual values. Deviations from the line indicate prediction errors. Patterns above or below the line may suggest systematic underprediction or overprediction in certain ranges.



6.5 Prediction Error Distribution

Visualization type: Residual histogram or error-distribution plot.

Residuals are defined as:

[

e_i=y_i-\hat{y}_i

]

What it shows:

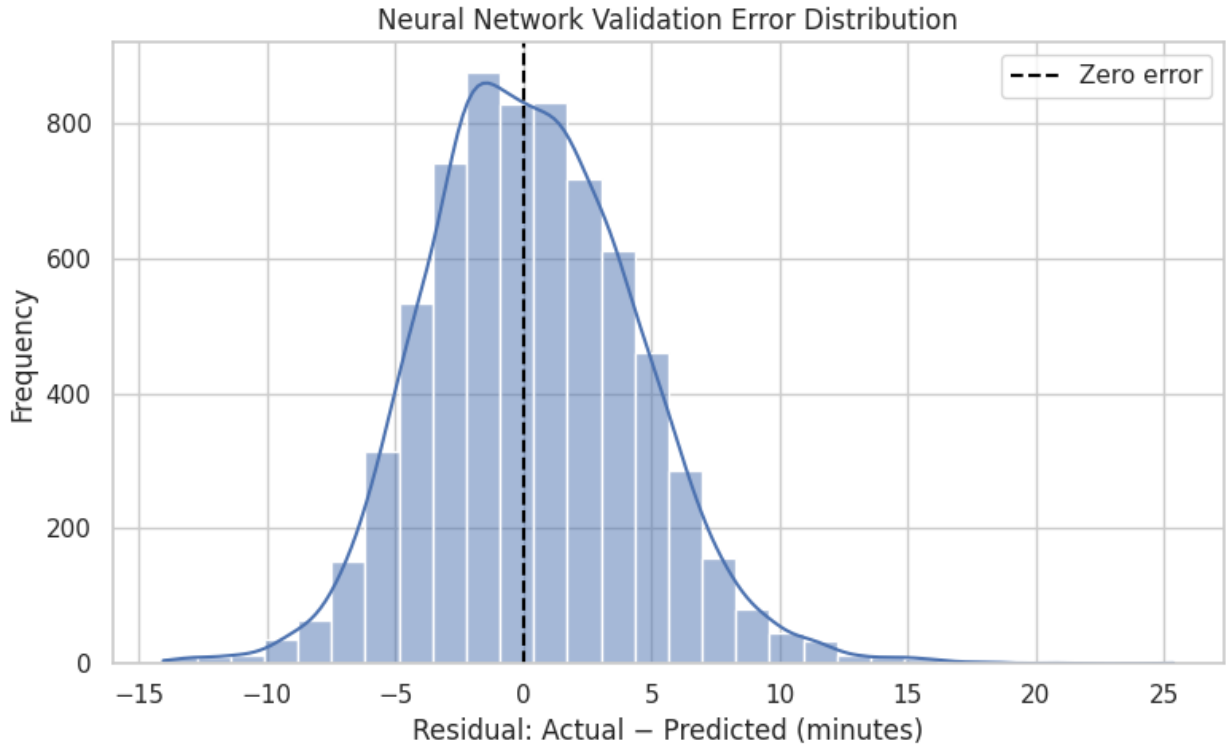
The distribution of residuals indicates the direction and magnitude of prediction errors.

Why it was created:

It helps determine whether the model tends to overpredict or underpredict delivery duration and whether large errors occur frequently.

Interpretation:

Residuals concentrated around zero indicate limited overall bias. A wide distribution or long tails indicate that some predictions have substantially larger errors.



6.6 Validation-Based Model Comparison

The baseline, Random Forest, and neural network were compared using validation metrics. The Random Forest was selected using validation MAE.

This approach separates model selection from final test evaluation and helps reduce the risk of reporting a test score that has been indirectly optimized through repeated model choices.

6.7 Final Test Results

The selected Random Forest model produced the following test-set results in the recorded notebook run:

Metric	Test result
Mean Absolute Error	3.16 minutes
Root Mean Squared Error	4.00 minutes
R^2 score	0.8140

Interpretation of results

The test MAE of 3.16 minutes means that the average absolute difference between predicted and actual delivery duration was approximately 3.16 minutes on the test set.

The RMSE of 4.00 minutes is higher than MAE, which is expected because RMSE gives greater weight to larger errors.

The R^2 score of 0.8140 indicates that the selected model explains approximately 81.4% of the observed variation in delivery duration relative to the mean-prediction baseline on this test split.

These results indicate that the selected model captures useful patterns in the dataset. However, individual predictions may still have larger errors, particularly for unusual delivery conditions or underrepresented groups.

7. FEATURE ANALYSIS

Feature analysis was performed to investigate which numerical variables had stronger linear relationships with delivery duration.

The notebook's numerical correlation analysis identified the following approximate relationships with the target:

Numerical feature	Pearson correlation with delivery time
`multiple_deliveries`	+0.387
`Delivery_person_Ratings`	-0.339
`Distance_km`	+0.322
`Delivery_person_Age`	+0.299
`Vehicle_condition`	-0.234
`Order_Hour`	+0.184

These values describe linear associations in the analyzed dataset.

The positive correlation for `multiple_deliveries` suggests that higher multiple-delivery categories tend to be associated with longer delivery durations. Distance also shows a positive association with the target. Delivery-person ratings and vehicle condition show negative correlations in this dataset.

Correlation is not equivalent to feature importance. It does not establish causation and does not capture all nonlinear relationships or interactions. Therefore, these results are used as exploratory evidence rather than definitive proof of the contribution of each feature to the trained model.

8. DISCUSSION

The project demonstrates how EDA and regression modeling can be combined to investigate a practical delivery-time prediction problem.

The target distribution showed that delivery duration is concentrated around the mid-twenties in minutes, with some longer deliveries. The IQR analysis identified a small proportion of potential outliers, which were retained because they may represent genuine operational delays.

The categorical analysis showed differences in observed delivery durations across traffic and weather categories. Grouped statistics also indicated differences by city, festival status, and multiple-delivery category. These observations may help guide further investigation of delivery operations.

The feature analysis identified multiple deliveries, ratings, estimated distance, and delivery-person age as numerical variables with comparatively stronger linear associations with delivery duration in this dataset.

The model evaluation showed that the Random Forest performed best according to validation MAE among the evaluated models. Its final test MAE was 3.16 minutes, RMSE was 4.00 minutes, and R^2 was 0.8140.

The results should be interpreted within the limitations of the available dataset. In particular, recorded categories and correlations do not demonstrate causation, and the model's performance on this dataset does not guarantee the same accuracy on new cities, future orders, or live operational data.

9. CHALLENGES AND LIMITATIONS

9.1 Missing and Inconsistent Values

Several fields contained missing values or textual placeholders representing missing data. Normalizing these values and applying appropriate imputation were necessary before modeling.

9.2 Geographical Distance Estimation

The calculated distance represents straight-line distance between restaurant and customer coordinates rather than actual travel distance. Real delivery routes may be longer due to road layouts, traffic restrictions, and route choices.

9.3 Outliers and Unusual Delivery Times

Some observations contained unusually long delivery durations. Removing all outliers could discard legitimate delayed orders, while retaining all of them can make model prediction more difficult.

9.4 Categorical Group Sizes

Some categories, particularly Semi-Urban city classification and bicycle deliveries, contained fewer records than the larger categories. Mean comparisons for smaller groups may be less stable.

9.5 Observational Data

The dataset supports association analysis but does not establish causation. For example, differences in delivery time between weather categories may also reflect distance, city, traffic, or other variables.

9.6 Model Generalization

The model was evaluated using a random holdout split from the available dataset. Its performance on future orders, different geographical regions, or changing operating conditions requires additional validation.

9.7 Prediction Error

Although the selected model achieved a useful test score, MAE and RMSE show that predictions are not exact. Some orders may have errors considerably larger than the average.

10. CONCLUSION

This project applied Exploratory Data Analysis, statistical summaries, visualization, feature engineering, preprocessing, and regression modeling to a real-world food delivery dataset containing 45,593 records.

The analysis investigated delivery-time distributions, potential outliers, estimated geographical distance, traffic density, weather conditions, vehicle types, city classification, festival status, and multiple deliveries. It also identified missing values and prepared numerical and categorical features for machine learning.

A Multilayer Perceptron neural network was developed using TensorFlow and Keras. Its performance was compared with a mean-prediction baseline and a Random Forest regressor. The Random Forest model was selected using validation MAE and achieved a test MAE of 3.16 minutes, RMSE of 4.00 minutes, and R^2 of 0.8140.

The findings indicate that data exploration and regression modeling can provide useful estimates of delivery duration. Nevertheless, observed relationships should not be interpreted as causal, and additional evaluation is needed before applying the model in a live delivery environment.

Overall, the project demonstrates the importance of data quality, meaningful visualizations, careful preprocessing, appropriate model selection, and independent test evaluation in solving a real-world prediction problem.

11. FUTURE SCOPE

The project can be extended in several ways.

1. **Use actual road distance and route information:** Integrate road-network distance, route duration, and route restrictions instead of relying only on straight-line distance.
2. **Include live traffic and weather:** Use time-stamped traffic and weather data to improve estimates under changing conditions.
3. **Perform systematic hyperparameter tuning:** Compare additional model configurations using a defined validation strategy.

4. **Explore additional regression models:** Evaluate gradient boosting, XGBoost or other suitable regression algorithms where available.
5. **Improve temporal validation:** Use chronological training and testing splits to examine performance on later orders.
6. **Analyze subgroup performance:** Compare prediction errors across cities, weather conditions, vehicle types, and delivery categories.
7. **Improve interpretability:** Use permutation importance or SHAP-based methods to examine model behavior more directly.
8. **Develop a user interface:** Build a simple application where users can enter delivery information and receive an estimated delivery duration.
9. **Monitor model performance:** If deployed, track prediction errors and retrain the model when the data distribution changes.
10. **Evaluate operational usefulness:** Study whether more accurate estimates improve customer communication and delivery planning.

12. REFERENCES

Use the following references for the tools and methods used in the project. Add the exact original dataset reference before submission.

1. **Dataset reference — required:**
   [Author or uploader]. ([Year]). *[Exact dataset title]* [Data set]. [Kaggle / UCI / other repository]. [Insert the exact dataset URL].
2. McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter* (3rd ed.). O'Reilly Media.
3. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., VanderPlas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html
4. Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering, 9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55
5. Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., Levenberg, J., Mané, D., Monga, R., Moore, S., Murray, D., Olah, C., Schuster, M., Shlens, J., Steiner, B., Sutskever, I., Talwar, K., Tucker, P., Vanhoucke, V., Vasudevan, V., Viégas, F., Vinyals, O., Warden, P., Wattenberg, M., Wicke, M., Yu, Y., & Zheng, X. (2016). TensorFlow: Large-scale

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13. APPENDIX

Appendix A: Source Code

The complete Python implementation is provided in the Jupyter Notebook:

`Food_Delivery_Delay_Analysis_Project.ipynb`

The notebook includes:

- Data loading and initial inspection.
- Missing-value and duplicate analysis.
- Data cleaning and feature engineering.
- Statistical summaries and EDA visualizations.
- Numerical and categorical preprocessing.
- Training, validation, and test splitting.
- Neural-network creation and training.
- Baseline and Random Forest model comparison.
- Final regression evaluation.
- Actual-versus-predicted and residual visualizations.

Appendix B: Software and Libraries

The project uses the following software and libraries:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- TensorFlow / Keras

Appendix C: Project Repository

GitHub repository: [https://github.com/anuskagupta123/food-delivery-delay-analysis.git]

The repository should include the project notebook, report in PDF or Word format, a README file, and either the dataset or its original source link.

Google Form submission: https://forms.gle/kT9DU5aERo2peTNh8